CAMBRA

Interventions / Treatments

CAMBRA Interventions

- Most of these are used to modify the biofilm
  - Make it less hospitable to acidophilic bacteria
- Some also kill bacteria

Regular Toothpaste

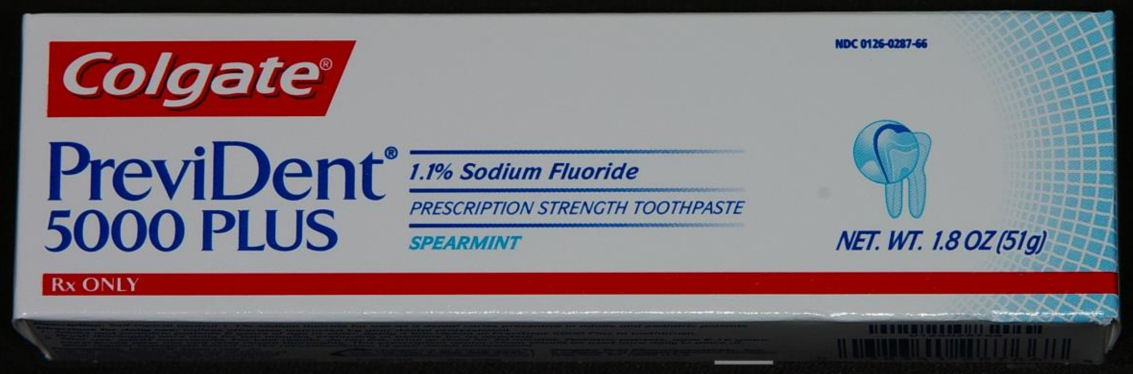
- For LOW and some MODERATE risk patients
- Toothpaste is milling paste – most will damage porcelain glaze and composite polish
  - Anything with an RDA > 80 is too abrasive
- Arm & Hammer Complete Care has an RDA of 35 – safe for all restorations
- Arm & Hammer also raises the pH of oral environment
  - Discourages acidophilic (cariogenic) bacteria

Photo from web

High Fluoride Toothpaste

Pea-size amount

Use baking soda with toothpaste (except Arm&Hammer). Sprinkle it on top of any toothpaste to raise pH.



Colgate Prevident 5000

Photo from web

Toothpaste is Milling Paste

too much toothpaste

Pea-size amount

Photo from web

Silver Diamine Fluoride

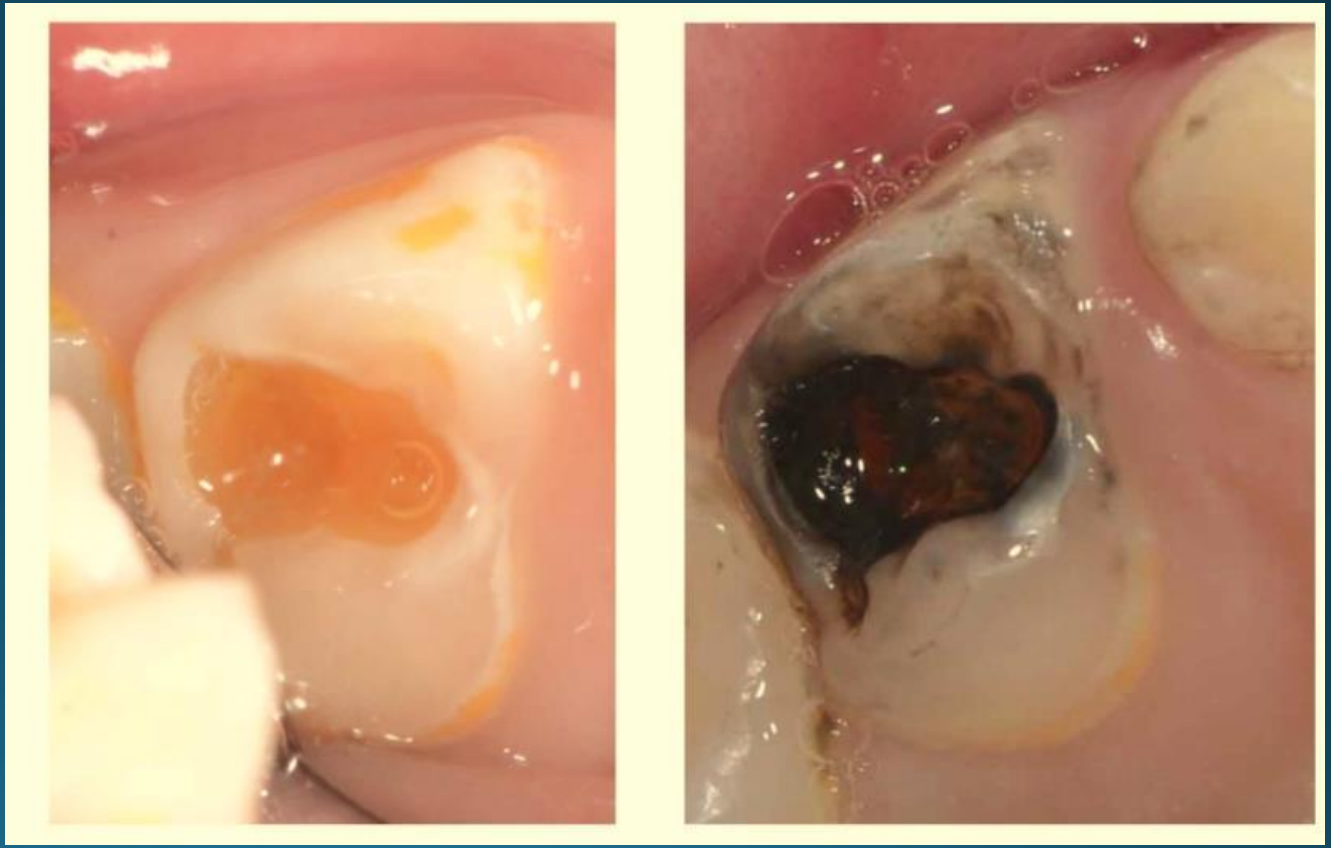


Photo from web

NO LONGER "OFF LABEL"

- The FDA granted Advantage Arrest Silver Diamine Fluoride a *Breakthrough Therapy Designation* on 30 October, 2016

- Use to arrest dental caries is no longer considered "off label use"

Redmond, OR and West Palm Beach, FL – October 30, 2016 -- Advantage Silver Dental Arrest, LLC and Elevate Oral Care, LLC announced today that the U.S. Food and Drug Administration (FDA) has granted *"Breakthrough Therapy Designation" to Advantage Arrest™ Silver Diamine Fluoride 38%* for the arrest of tooth decay in children and adults.

SILVER DIAMINE FLUORIDE

- 38% SFD solution contains
  - Silver ions – 253,870 ppm
  - Fluoride ions – 44,800 ppm

- The high F^+ concentration can be toxic when swallowed in <u>large</u> amounts

- Use only one drop per treatment session and <u>RINSE</u>

Mei, et al; antibacterial effects of silver diamine fluoride on multi-species cariogenic biofilm on caries

SILVER DIAMINE FLUORIDE

- Silver ions inactivate glucosyltransferase enzymes which stops bacterial glucan production
  - Prevents bacterial adhesion to tooth
- Silver is toxic to bacteria
- Fluoride ions bind to enolase and ATPase effectively stopping bacterial carbohydrate metabolism of acidogenic oral bacteria
  - Kills bacteria
- Fluoride strengthen enamel by forming fluorapatite

SILVER DIAMINE FLUORIDE

- Since ancient times, the silver ion has been known to be effective against a broad range of microorganisms
- Interaction of silver ions with thiol groups in enzymes and proteins plays an essential role in its antimicrobial action
- Between 2.5 and 5 ppm Ag is toxic to bacteria <u>and</u> human mesenchymal cells

W.K. Jung, et. Al., Appl Environ Microbiol. 2008 Apr; 74(7): 2171–2178

C. Greulich, et.al., RSC Advances, 2012, 2, 6981–6987

EFFECTS ON BONDING

- 50% reduction in GIC bond strength but rinsing with water restores bond strength
- 33% reduction in resin bond strength but again, rinsing restores the bond strength
- SDF inhibits matrix metalloproteases in dentin collagen

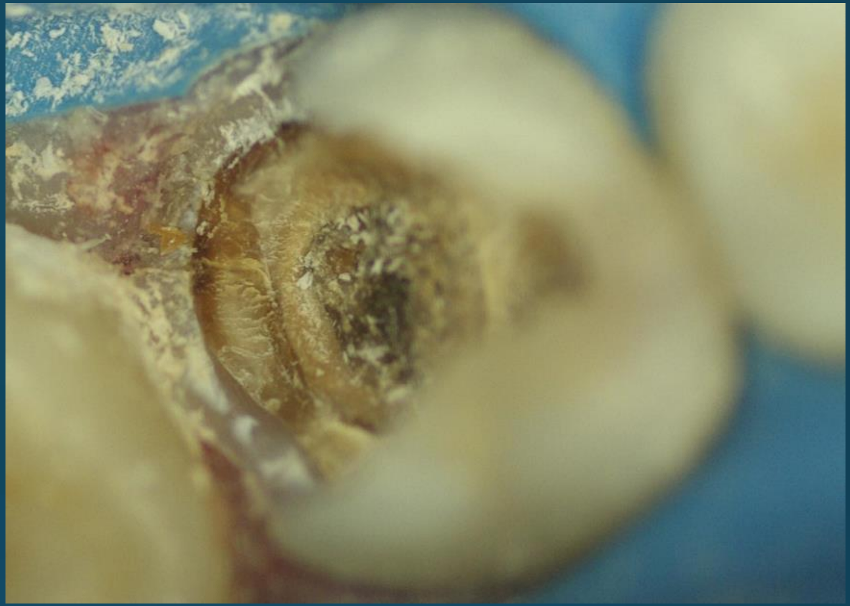
Knight, et.al., Aust Dent J 2006;51(1):42-45

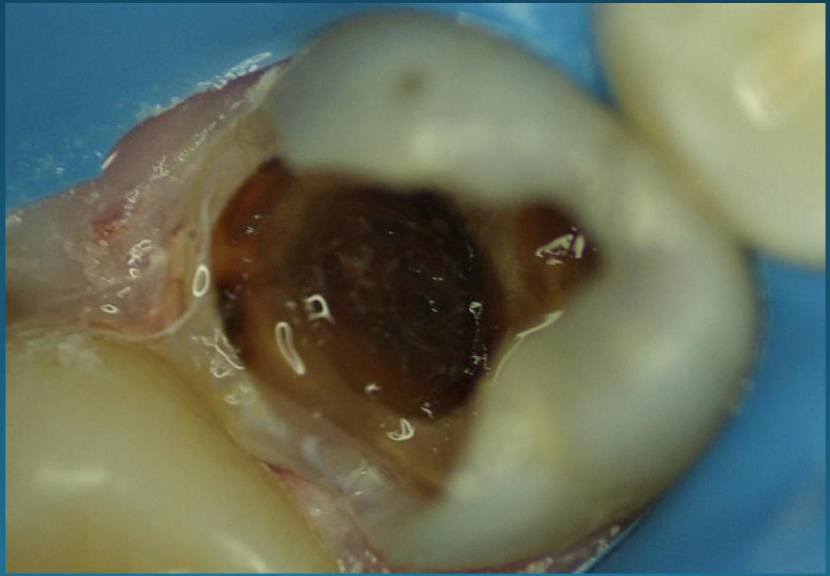
Mei, et al; antibacterial effects of silver diamine fluoride on multi-species cariogenic biofilm on caries

HOW TO APPLY

- Allow SDF to soak into and react with the lesion for 1 to 3 minutes.
- Then rinse with <u>copious</u> water and either leave exposed or cover with GIC

SDF in Action





SDF in Action

M.O. Coover, DDS, MAGD

Indications:

1. Extreme caries risk (xerostomia or severe early childhood caries).
2. Treatment challenged by behavioral or medical management.
3. Patients with carious lesions that may not all be treated in one or two visits.
4. Difficult to treat dental carious lesions.
5. Patients without access to dental care.
6. Cervical hypersensitivity ☼
   No, it doesn't stain non-carious teeth

MAXIMUM DOSE

- Maximum dose: 25 µL (1 drop) / 10kg per treatment visit.
- SDF Contraindication: Silver allergy.
- SDF Relative Contra-indications: Ulcerative gingivitis, stomatitis.
- SS-KI Contraindications: Pregnancy, breastfeeding.

CONSIDERATIONS:

- Decayed dentin will darken as the caries lesions arrest. Most will be dark brown or black.
- SDF can stain the skin, which will clear in two to three weeks without treatment.
- SDF can **permanently stain operatory surfaces and clothes.**

CONSIDERATIONS:

- A provisional restoration (e.g., GI via ART or other material) may be considered after SDF treatment. [remember to rinse first]
- Saturated solution of potassium iodide (SS-KI, [not Lugol's Solution]) can be used after SDF to decrease color changes (but not much).
- Re-application is usually recommended, biannually until the cavity is restored or arrested or the tooth exfoliates.

SDF Cost

Advantage Arrest Silver Diamine Fluoride 38% - Unit-Dose Ampule

Box of 30 Ampules plus 30 each small and large applicators

Quantity	Price
1	$119.95
2	$109.95
3 +	$99.95

1 Add to Shopping Cart

Advantage Arrest Silver Diamine Fluoride 38% - Bottle

Each bottle contains 8 mL

Quantity	Price
1	$149.00
2	$141.50
3 – 11	$134.50
12 +	$119.00

8 ml is enough for 160 patients

SDF Cost

- Use 1 drop per patient
  - 1 drop treats 3-5 sites
- Bottle contains 8ml = 160 drops
- $149 ÷ 160 = 93¢ per drop

How to Get Paid

- Code **D1354**: Interim caries arresting medicament application;
- Code D1208: Topical application of fluoride;
- Code D9910: Application of a desensitizing medicament, per visit; or
- Code D1999: Unspecified preventive procedure by report.

- You may have to identify your patient's caries risk to justify reimbursement with a recognized caries risk tool. Codes D0601 (low risk), D0602 (moderate risk), D0603 (high risk) are especially helpful in all claims in order to measure the change in caries risk of the patients in your practice.

Xylitol


Xylishield
xylitol mints
ULTRADENT


Epic
45% more xylitol
than other brands
Spearmint Gum
100%
Xylitol
SWEETENED
12 Pieces
The Cavity-Frightening, Naturally-Sweetened, Great-Tasting, Sugar-Free,
Xylitol-Powered, Kid-Safe, Non-GMO, Gluten-Free, Lots-of-Hyphens kind of gum

Xylitol

- **Five carbon sugar alcohol**
- **Not fermentable by mutans strep.**
- **Chew 2 gms for five minutes 3-5 times per day**

"These results and previous studies suggest that high xylitol chewing gum usage can retard or arrest even rampant dentine caries in conditions where effective restoration and prevention programs have not been instituted, and can also provide additional protection against further caries development during full implementation of restorative procedures by holding the lesion in a non-progressive condition."

Makinen KK et al
1 Suppl 1
International Dental Journal 1995

How does Xylitol work?

- **Starvation Effect**
- **Environmental Effect**
- **Adhesion Effect**

Starvation Effect

- S. mutans readily absorbs xylitol
- S. mutans cannot metabolize xylitol, the bacteria that have absorbed xylitol starve

Environmental Effect

- S. mutans is acidophilic and acidogenic
- As some of the S. mutans starve and die, the plaque biofilm becomes less acidic, less hospitable to acidophilic bacteria.
- The biome slowly shifts to less cariogenic bacteria

Adhesion Effect

- As the plaque biome shifts, the replacement bacteria, while still acidogenic, are less able to adhere to tooth surfaces long enough to foster demineralization and caries .

MAINTAINING MUTANS STREPTOCOCCI SUPPRESSION WITH XYLITOL CHEWING GUM

Subjects rinsed with 0.12 percent CHX gluconate mouthrinse twice daily for 14 days.

Those in the test group (n = 51) chewed a commercial xylitol gum three times daily for a minimum of five minutes each time for three months. The placebo group subjects (n = 50) used a commercial sorbitol gum, and the control group subjects (n = 50) did not chew gum.

Hildebrandt Gh, Sparks BS, J Am Dent Assoc, 2000; 131 (JUly): 909-916

"At 3 months after CHX rinsing, mean MS levels increased toward baseline but at different rates. The placebo group increased 40-fold, the control group increased 25-fold, and the test group had an 8-fold increase…"

Conclusion: The use of xylitol chewing gum significantly delays the return of oral MS after therapeutic use of a CHX mouthrinse.

Maintaining Mutans Streptococci Suppression With Xylitol Chewing Gum

Maintaining Mutans Streptococci Suppression With Xylitol Chewing Gum
Hildebrandt Gh, Sparks BS, J Am Dent Assoc, 2000; 131 (JUly): 909-916

A Word about Dogs

GCH Geordi, CGC, CGC-U, CGC-A

In humans, Xylitol is absorbed slowly and has no measurable effect on insulin production.

Unfortunately, the same cannot be said about dogs. **When dogs eat Xylitol, their bodies mistakenly think that they've ingested a huge amount of glucose and start producing large amounts of insulin.**

When this happen, the dog's cells start rapidly taking up glucose from the bloodstream. This can lead to hypoglycemia (low blood sugar levels) and is fatal if not treated immediately.

Xylitol may also have detrimental effects on liver function in dogs, with high doses causing liver failure.

So if you own a dog, then keep Xylitol out of reach (or out of your house altogether). If you believe your dog accidentally ate Xylitol, take it to the vet <u>immediately</u>.

Xylitol Occurs Naturally

- Xylitol is found naturally in a number of fruits and vegetables (including strawberries, raspberries, pears, cauliflower and plums).
- The human body produces between 5 and 15 grams of xylitol each day as it metabolizes carbohydrates. (It's an intermediate compound in the glucuronate-xyulose metabolic pathway.)
- When manufactured, the starting agricultural product is usually birch trees or corn cobs.

Xylitol Dosage

- 1) Six to ten grams per day.
- 2) Divided up into three to five doses throughout the day.
- 3) Continuation of this regimen for six months to a year before it can be assumed that maximum anti-cavity protection exists.

Dosage

- The existence of a dosing ceiling was confirmed by a study [Milgrom, 2006] that evaluated different xylitol doses in regard to their ability to create changes in the level of cariogenic bacteria found in dental plaque and saliva.
- **Amounts exceeding 10.3 grams per day were not likely to increase effectiveness.**
- **Doses of 3.5 grams or less per day were ineffective**
- **2 pieces gum or mints, 3 tid = 6 grams**

Dry Mouth

- Biotene has long been used as a saliva substitute. It is useful during the day, especially in this spray form, but it does little at night.

Photo from Web

Dry Mouth

Spry makes a saliva substitute that contains xylitol.

Photo from Web

Dry Mouth

XyliMelts stimulate saliva production at night and also contain xylitol.

Photo from Web

Anti-Microbials

- Typically used only is cases of extreme rampant caries

Photo from Web

Chlorhexidine

- Dosage for Caries control
  - Rinse with ½ ounce once per day for seven days. Repeat monthly.
  - One bottle lasts four months; rarely used beyond that.
- Chlorhexidine binds to fluoride so use 2 hours before or two hours after brushing.

Photo from Web

Povidone Iodine

- Iodine is another anti-microbial used to re-set the oral microbiome to allow normal oral bacteria to re-populate the mouth. It is applied in-office once per week using a cotton-tip applicator to paint it on the teeth once a week for two weeks.
- Brian Novi used it as a one-time rinse but this is very messy and patients don’t much like it.

Photo from Web

Fluoride Rinses

Fluoride Rinses are used mid-day after lunch, rinse vigorously and spit.

Synedent fluoride is available from Amazon.com. It contains 250 ppm fluoride, chitosan-argininamide. Arginine is metabolized to NH_4 by oral bacteria thus raising the oral pH. This is currently not on the market.

ACT is available in any drug or grocery store. 250 ppm fluoride

Photo from Web

Fluoride Varnish

- Moderate and High-Risk patients should have fluoride varnish applied to the teeth at every prophy.

Photo from Web

Pit & Fissure Sealants

- 90% retention at 10 years (if done correctly)
  - Correct bonding and isolation are critical
    - Biofilm removal
    - Etch
    - Prime
    - Bond
    - Apply sealant sparingly
- Insurance will not cover sealants in adults unless there is proof that patient is High Caries Risk
  - Always do a Caries Risk Assessment

Photo from Web

Probiotics

- This particular pro-biotic has been developed by dentists (Dr. Jeffery Hillman) to re-populate the mouth with "normal" oral bacteria to change the biome of disease to one approaching normal.
- Dr. Dan Sneed was an early proponent and owned stock in the company that developed it.

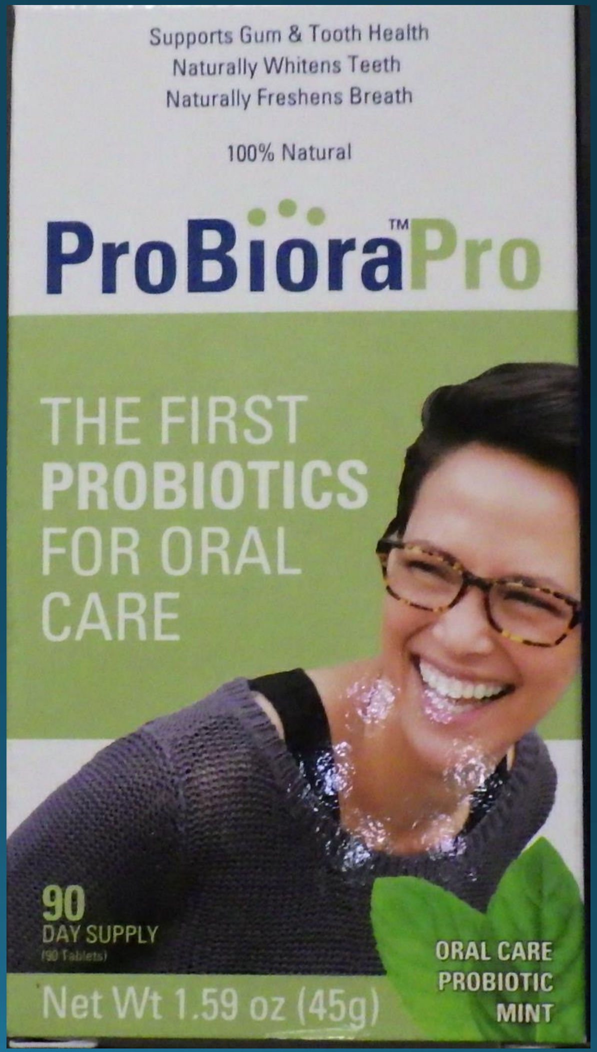


Photo from Web

Patient Recall Interval

- Low Risk – every 6 months
- Moderate Risk – every 4-6 months
- High Risk – every 3-4 months
- Extreme Risk – every 2-4 months

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The End